

①

Stack (LIFO) - Array size = 5

operations : push(45), push(22), push(11), push(77), pop(),
push(44), push(33), pop(), pop(), pop()

operation	stack (bottom → Top)	Top
		0
push(45)	45	
push(22)	45, 22	
push(11)	45, 22, 11	2
push(77)	45, 22, 11, 77	3
pop()	45, 22, 11	2
push(44)	45, 22, 11, 44	3
push(33)	45, 22, 11, 44, 33	4
pop()	45, 22, 11, 44	3
pop()	45, 22, 11	2
pop()	45, 22	1

Peak element = 22, Index position = 1

LIFO principle = Last in first out.

②

Circular Queue (size = 5)

step	operation	Queue
		20
1.	Enqueue(20)	20, 10
2.	Enqueue(20)	20, 10, 20
3.	Enqueue(30)	10, 30
4.	Dequeue()	10, 30, 20
5.	Enqueue()	
6.	Dequeue()	30, 20

Analytical problems.

step	operation	queue
7.	Enqueue (90)	30, 80, 90
8.	Enqueue (100)	30, 80, 90, 100
9.	Dequeue ()	80, 90, 100
10.	Dequeue ()	90, 100
11.	Enqueue (60)	90, 100, 60
12.	Enqueue (90)	90, 100, 60, 90

Array position (0 to 4)

Index $\Rightarrow$ 0 = 60

Index $\Rightarrow$ 1 = 90 (rear)

Index $\Rightarrow$ 2 = 90 (front)

Index $\Rightarrow$ 3 = 100

Index $\Rightarrow$ 4 = Empty

Front Element = 90 (position / index = 2)

Rear Element = 90 (position / index = 1)

③ Binary Search

Elements : 66, 111, 212, 323, 432, 543, 643, 649, 777, 888

Case 1 : Key = 111

Element found at position = 10 (Index = 9)

Case 2 :-

Key = 888

element found at position = 10 (Index = 9)

Case 3 : Key = 999

Element not found

Time complexity : $O(\log n)$

Infix to postfix :-

Expression : $B - (C/D + (E/F * G)/H) * J$

postfix Expression
 $B C D / E F / G * H / + J * -$

Evaluate postfix

a) $9234 / * +$

Result = 15

b) $62 + 63 / *$

Result = 16

Queue operation (size = 5)

operation

Queue

Enqueue (21)

21

Enqueue (67)

21, 67

Enqueue (89)

21, 67, 89

Dequeue ()

67, 89

Enqueue (15)

67, 89, 15

Enqueue (30)

67, 89, 15, 30

Enqueue (52)

67, 89, 15, 30, 52

Dequeue ()

89, 15, 30, 52

Dequeue ()

15, 30, 52

Dequeue ()

30, 52

Dequeue ()

52

Final Queue :

Front = rear = 52

Final Queue $\rightarrow$ front = rear = 52

⑥ Balanced parenthesis :-

i) $[9 * (x + 8 - \{6 + e\} + 77)] / 100$

Not Balanced

ii) $[(a(a * b) + (c/2) - d)]$

Not Balanced

⑦ prefix and postfix

Expression :

$$(a+b) + c * (d+e) + f)^* (g+h)$$

prefix :

$$* + + + a b * c + d e f + g h$$

postfix :

$$a b + c d e + * + f + g + h f *$$

Final Queue $\rightarrow$ front = rear = 52

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Not Balanced

ii) $[(a(a*b) + (c/2) - d)]$

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